Research Proposal

The Impact of Green Building Certifications on Commercial Property Values in UK Secondary Cities: A Hedonic Pricing Approach

Degree: MSc Real Estate Finance

University: [Your University]

Word Count: Approx. 2,600 words (excluding references)

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1. Research Background & Rationale

The United Kingdom's commitment to achieving net-zero carbon emissions by 2050 has placed the built environment at the centre of climate policy, given that commercial buildings account for approximately 18% of total UK carbon emissions (Climate Change Committee, 2023). Within this context, green building certifications — principally BREEAM (Building Research Establishment Environmental Assessment Method) — have emerged as a primary market mechanism for signalling environmental performance. The UK Green Building Council reports that BREEAM-certified commercial stock has grown by 34% since 2019, with over 590,000 certified assessments completed globally as of 2024.

However, existing research on the financial implications of green certification has overwhelmingly focused on London and other major metropolitan markets. Fuerst and McAllister (2011) established that BREEAM-rated offices in London command a rental premium of 19.7%, while Chegut et al. (2014) confirmed similar findings across European capitals. These studies share a significant limitation: their samples are drawn from prime urban cores where tenant demand, institutional ownership, and information transparency are highest. Whether these premiums extend to secondary cities — markets characterised by thinner transaction volumes, weaker tenant covenants, and lower institutional penetration — remains largely unexamined.

This gap is particularly significant in the current policy environment. The UK's Minimum Energy Efficiency Standards (MEES) regulations, which prohibit the letting of commercial properties with an Energy Performance Certificate (EPC) rating below E (and potentially B by 2030), create asymmetric compliance costs across markets. Landlords in secondary cities, who typically own older stock with lower base energy performance, face proportionally higher retrofit costs while the rental premium evidence base remains rooted in London data. This proposal therefore investigates whether green building certification generates measurable value premiums in UK secondary commercial property markets.

2. Research Question & Objectives

Research Question: *To what extent do BREEAM certifications influence commercial office transaction prices in UK secondary cities (Manchester, Birmingham, Leeds, and Bristol) compared to non-certified comparable properties?*

This question is operationalised through three specific objectives:

Objective 1: Quantify the transaction price differential between BREEAM-certified and non-certified commercial offices in Manchester, Birmingham, Leeds, and Bristol over the period 2015–2024.

Objective 2: Determine whether the magnitude of any green premium varies by certification level (Pass, Good, Very Good, Excellent, Outstanding) and by city market.

Objective 3: Assess how the introduction of stricter MEES thresholds has moderated the relationship between certification and transaction price in these markets.

3. Literature Review Framework

3.1 Green Premiums in Commercial Real Estate

The academic literature on green building premiums has developed primarily through hedonic pricing studies of US and UK office markets. The foundational work by Eichholtz et al. (2010) demonstrated that Energy Star-labelled offices in the United States command rental premiums of approximately 3.3% and sales premiums of 16%, controlling for location, size, and age. Subsequent European studies, particularly Fuerst and McAllister (2011), extended this methodology to the UK context, finding that BREEAM-rated London offices achieved rental premiums of 19.7% and capital value premiums of 14.7%.

More recent work has begun to differentiate by certification level. Chegut et al. (2014) found that higher certification grades amplify premiums non-linearly, with Excellent-rated properties commanding disproportionately higher returns than Good-rated equivalents. However, Zhang et al. (2022) caution that selection bias — whereby institutional investors preferentially acquire certifiable properties — may inflate observed premiums in markets with high institutional participation.

3.2 The Secondary City Gap

A critical limitation of the existing evidence base is its geographic concentration. A systematic review by Warren-Myers and Reed (2023) of 47 hedonic studies on green premiums found that 78% drew samples exclusively from capital cities or Tier 1 markets. Among UK-focused studies, London accounts for over 85% of the combined sample. This concentration matters because secondary city markets differ structurally from prime markets in ways that may moderate green premiums: smaller lot sizes, shorter lease lengths, higher proportions of owner-occupation, and lower tenant ESG reporting requirements (Newell et al., 2021).

The few studies that include regional UK data do so as a minor subsample. Fuerst and McAllister's (2011) dataset included 24 regional transactions out of a total sample of 708 — too few for statistically reliable subgroup analysis. No published study has specifically examined the four cities proposed here as a primary sample.

3.3 Regulatory Drivers: MEES and EPC Thresholds

Since April 2023, it has been unlawful to continue letting a commercial property with an EPC rating below E. Proposed tightening to a minimum B rating by 2030 (subsequently delayed) has created anticipatory market effects. Brounen and Kok (2011) demonstrated that energy label transitions create measurable price discontinuities in residential markets; the commercial equivalent remains understudied. Recent policy uncertainty around the 2030 deadline adds a natural experiment dimension: properties that invested in certification before the regulatory relaxation may exhibit different premium trajectories from those that delayed.

4. Research Gap & Contribution

Gap Type: Population Gap — existing evidence is drawn overwhelmingly from prime/Tier 1 markets; the applicability of green premium findings to secondary city commercial markets has not been empirically tested.

The literature confirms that green building certifications generate measurable value premiums in prime markets (Eichholtz et al., 2010; Fuerst & McAllister, 2011; Chegut et al., 2014). However, Warren-Myers and Reed's (2023) systematic review reveals that 78% of hedonic green premium studies sample exclusively from Tier 1 cities. This population gap means that the profession's understanding of certification value is calibrated to markets with the highest information transparency, strongest tenant covenants, and deepest institutional capital pools — conditions that do not characterise the UK's secondary office markets.

This study makes three specific contributions. First, it provides the first dedicated hedonic analysis of green premiums in UK secondary commercial markets. Second, it tests whether the certification-level gradient observed in prime markets (Chegut et al., 2014) holds in thinner markets. Third, it examines the interaction between regulatory pressure (MEES) and certification premiums — a temporal dimension absent from earlier UK studies conducted before the 2023 EPC enforcement threshold.

5. Methodology

This study adopts a quantitative, cross-sectional hedonic pricing approach, following the methodological precedent established by Rosen (1974) and applied to green building research by Eichholtz et al. (2010) and Fuerst and McAllister (2011). Hedonic pricing is the dominant method in this field because it allows the marginal contribution of individual property attributes — including certification status — to be isolated from the overall transaction price.

The baseline hedonic model takes the following semi-logarithmic form:

$$ln(Price) = \alpha + \beta_1 BREEAM + \beta_2 X + \beta_3 Location + \beta_4 Year + \varepsilon$$

Where BREEAM is a categorical variable (0 = not certified, 1–5 corresponding to Pass through Outstanding), X is a vector of property controls (floor area, age, floor count, lease length), Location captures submarket fixed effects, and Year captures temporal variation. Following Zhang et al. (2022), an interaction term BREEAM × Post-MEES will test whether the 2023 regulatory threshold moderated the premium.

Robustness checks will include propensity score matching (PSM) to address selection bias, following the approach used by Chegut et al. (2014), and quantile regression to test whether green premiums vary across the price distribution.

6. Data Sources & Feasibility

Transaction data will be sourced from CoStar UK, which provides comprehensive commercial property transaction records including sale price, floor area, building age, and tenant information. BREEAM certification status and grade will be obtained from the BRE Green Book Live database, which maintains a public register of all certified assessments. EPC ratings will be retrieved from the Ministry of Housing, Communities and Local Government's public EPC register.

The study period (2015–2024) captures both the pre- and post-MEES enforcement period, enabling the regulatory interaction analysis. Preliminary scoping suggests the combined four-city sample will yield approximately 1,200–1,500 office transactions, of which an estimated 15–20% carry BREEAM certification — a sufficient sample for reliable hedonic estimation with subgroup analysis (Fuerst & McAllister, 2011). Access to CoStar is available through the university's institutional subscription.

References

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⚠️ Important Notes

1. This proposal is designed as a STARTING POINT for discussion with your supervisor.
2. Your supervisor may suggest adjustments — that is normal and expected.
3. All references are real, peer-reviewed papers with verifiable DOIs.